

1. 2013Q2

Question 2 refers to the table below, which shows the relative amounts of mitochondria, chloroplasts and endoplasmic reticulum in four cell types P, Q, R and S:

Cell type	Relative amount of the sub-cellular structure		
	Mitochondrion	Chloroplast	Endoplasmic reticulum
P	+	++	+
Q	+++	-	+
R	+++	-	+++
S	+	-	+

Key: number of '+' indicates the relative amount of the sub-cellular structure
'-' indicates the absence of the sub-cellular structure

Which cell type would you expect to be insulin-producing cells in the pancreas?

- A. P
- B. Q
- C. R
- D. S

2. 2013Q33

A person has not taken any food for 24 hours. Which of the following will increase in concentration in the blood?

- A. glucagon
- B. glucose
- C. glycogen
- D. insulin

3. 2013Q34

Which of the following information concerning the characteristics of insulin-dependent diabetes and non-insulin-dependent diabetes is correct?

<i>Insulin-dependent diabetes</i>	<i>Non-insulin-dependent diabetes</i>
A. accounts for the majority of diabetic cases	accounts for a small proportion of diabetic cases
B. is mainly due to an unhealthy lifestyle	is mainly due to hereditary factors
C. requires regular injections of insulin	may be controlled through proper diet
D. body fails to respond to insulin	body produces a low level of insulin

4. 2017Q36

Which of the following combinations correctly matches the type of diabetes with its description?

<i>Type of diabetes</i>	<i>Description</i>
A. Insulin-dependent (type I)	Heredity is the major cause.
B. Insulin-dependent (type I)	Blood insulin level remains high.
C. Non-insulin-dependent (type II)	Blood insulin level remains low even after meal.
D. Non-insulin-dependent (type II)	Blood glucose level will drop significantly after insulin injection.

5. 2019Q35

Type II diabetic patients may feel dizzy after prolonged exercise. Which of the following is the most likely explanation for this?

- A. Their blood glucose level has dropped to too low a level because they do not have enough stored glycogen to replenish the glucose used.
- B. Their blood glucose level has dropped to too low a level because they do not have enough glucagon to stimulate the conversion of glycogen to glucose.
- C. Their blood glucose level has risen to too high a level because they keep losing water during exercise.
- D. Their blood glucose level has risen to too high a level because they do not have enough insulin to stimulate the conversion of glucose to glycogen.

6. 2020Q9

The electron micrograph below shows an organelle P:

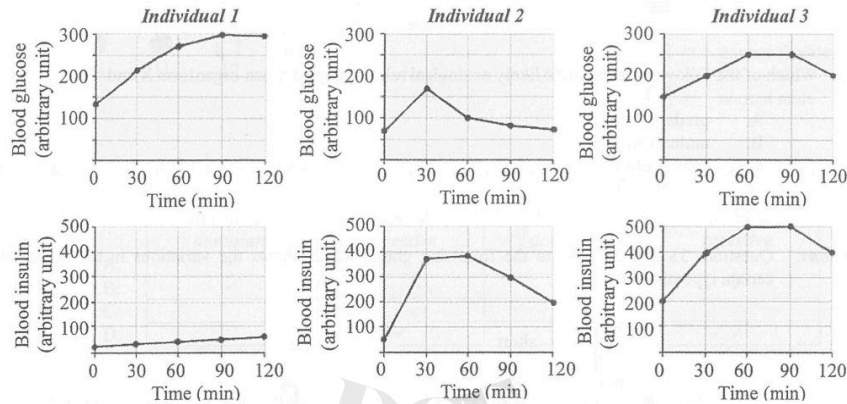


Which of the following are possibly produced by organelle P in the cells of the pancreas?

- (1) amylase
- (2) insulin
- (3) lipase
- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

7. 2020Q31

The graphs below show the changes in blood glucose level and the blood insulin level of three individuals after consuming a sugary drink:

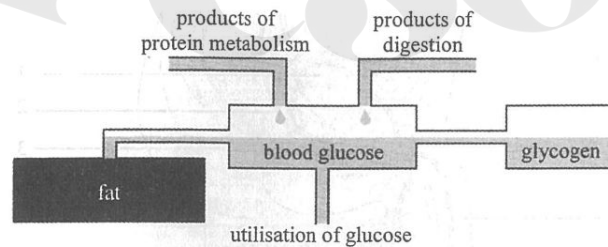


Which of the following combinations correctly shows the health conditions of these three individuals?

- | | Individual 1 | Individual 2 | Individual 3 |
|----|-----------------|-----------------|-----------------|
| A. | No diabetes | Type 1 diabetes | Type 2 diabetes |
| B. | Type 1 diabetes | Type 2 diabetes | No diabetes |
| C. | Type 2 diabetes | No diabetes | Type 1 diabetes |
| D. | Type 1 diabetes | No diabetes | Type 2 diabetes |

8. 2021Q26

Question 26 refer to the model below, which shows the regulation of blood glucose level in humans:

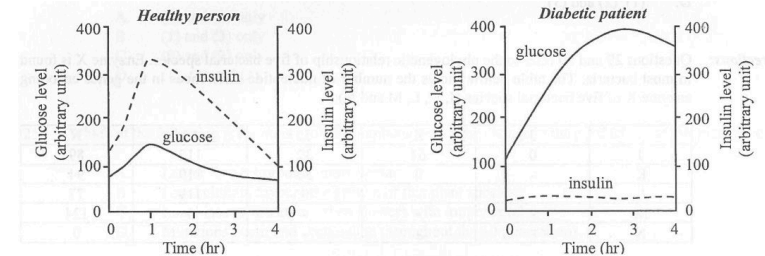


A student adds the following remarks to this model. Which remark is *incorrect*?

- Glycogen can be stored in muscles.
- Insulin converts glucose to glycogen for storage.
- The products of digestion take the form of simple sugars.
- The products of protein metabolism come from the liver.

9. 2023Q32

Questions 32 and 33 refer to the graphs below, which show the changes in the blood glucose levels and blood insulin levels of a healthy person and a diabetic patient after consuming a sugary drink:



Based on the graphs, which of the following combinations correctly matches the type of diabetes the patient is suffering from and its explanation?

- | Diabetes | Explanation |
|-----------|---|
| A. type 1 | blood glucose level remains high after the patient consumes a sugary drink |
| B. type 1 | insulin level shows no response to the rise in blood glucose level |
| C. type 2 | blood glucose level has a sharp rise after the patient consuming a sugary drink |
| D. type | liver fails to convert glucose to glycogen when blood glucose level is high |

10. 2023Q33

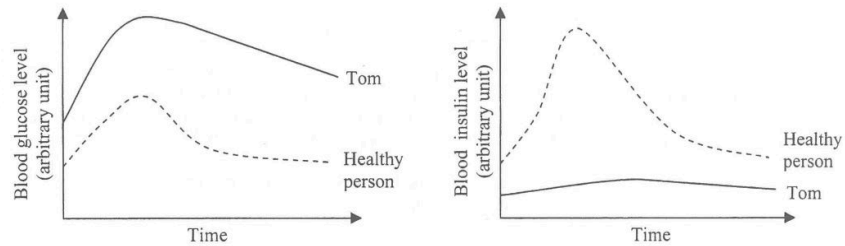
The blood glucose level of the patient eventually drops because of

- the actions of glucagon.
 - respiration of body cells.
 - loss of glucose through urine.
- (3) only
 - (1) and (2) only
 - (2) and (3) only
 - (1), (2) and (3)

LQ

1. 2012Q7

Tom suffers from diabetes. His doctor asked him to drink, after overnight fasting, a large volume of a glucose solution. After that, blood samples were taken at regular time intervals to measure insulin and glucose contents. The following graphs show the changes in Tom's blood glucose level and blood insulin level after the test, and those of a healthy person.



- Which type of diabetes does Tom suffer from? Explain your answer. (4 marks)
- Explain the difference in blood glucose response to the oral consumption of glucose solution between Tom and the healthy person. (3 marks)
- What medical treatment should Tom be given? (1 mark)

2. 2015Q8

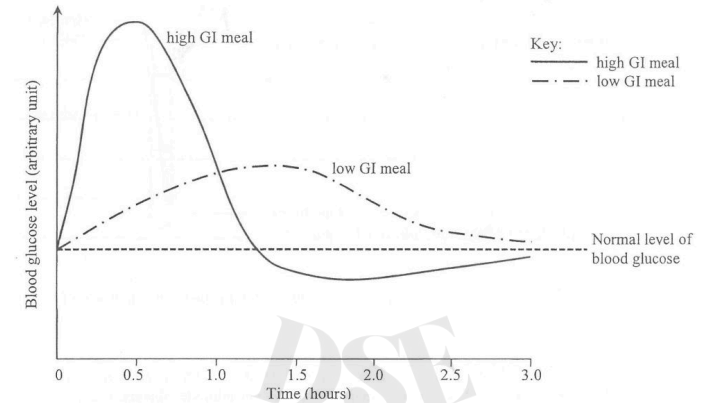
In a health check, Lisa was found to have glucose in her urine. She undertook a further check in which she has fasted for 12 hours before a blood sample was taken for examination. The results of the blood tests are shown below:

Tests	Results	Normal range	Units
Blood glucose	8.4	4-6	mmol L ⁻¹
Insulin	0.2	3-32	μU mL ⁻¹
Glucagon	130	20-100	μg L ⁻¹

- State the type of diabetes Lisa is suffering from. (1 mark)
- With reference to the production and actions of the two hormones, account for the results of Lisa's blood tests.
 - Insulin (3 marks)
 - Glucagon (3 marks)
- Suggest **two** dietary habits that Lisa should establish. (2 marks)

3. 2018Q7

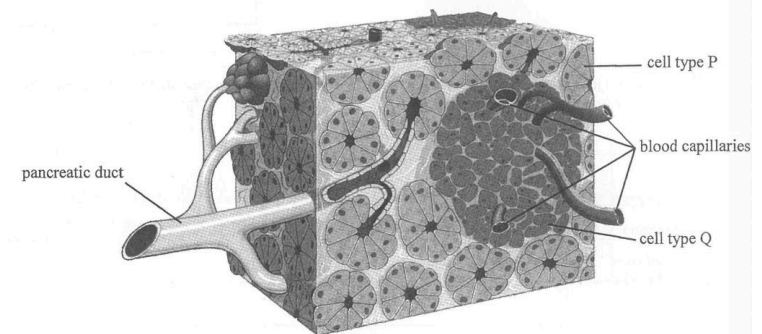
The glycemic index (GI) is an indication of the effect of food on the blood glucose level. The higher the GI value of a food, the quicker is the rise in blood glucose level. The graph below shows the changes in the blood glucose level of a healthy individual after consuming the same quantity of a low GI or high GI meal over a period of three hours.



- Describe how the consumption of a meal leads to an increase in the blood glucose level. (2 marks)
- Name the key hormone which lowers the blood glucose level. (1 mark)
 - Describe how this hormone lowers the blood glucose level. (3 marks)
 - On the graph, sketch a curve to show the change in the level of this key hormone in response to the consumption of a high GI meal by the healthy person. (2 marks)
- Explain why diabetic patients should consume low GI meals. (2 marks)

4. 2022Q6b

The schematic diagram below illustrates the distribution of different cell types in the human pancreas:



- A person jogs slowly for an hour. Describe how the hormones from the pancreas can regulate the blood glucose level of the person while jogging. (4 marks)

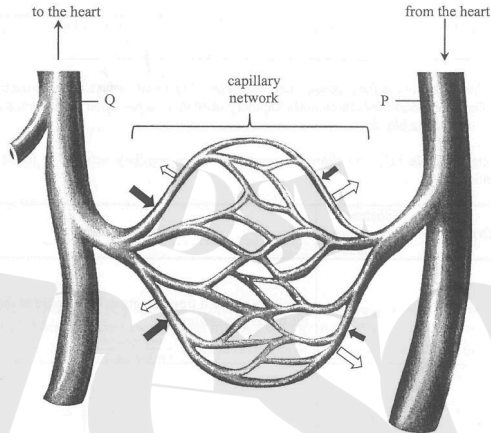
5. 2022Q11 - Essay

Recently, the use of the ketogenic diet for achieving weight loss is becoming popular. In fact, this high-fat, moderate-protein and very-low-carbohydrate diet has been used as an approach to control the blood glucose level in diabetics. However, the effectiveness of this diet in achieving weight loss is still controversial.

Describe how a ketogenic diet can be used to control the blood glucose level in diabetics. Evaluate the possibility of using this diet for weight loss and discuss the health concerns of adopting such a diet for healthy persons. (12 marks)

6. 2024Q4c

The schematic diagram below shows the arrangements of some blood vessels:



c) The capillary network is the location where exchange of materials occurs between the blood and tissue fluid. When the blood flows through the capillary network of a particular organ, some substances will be taken up into the blood. (3 marks)

Complete the table below to show the organ where the capillary network is found. Provide your explanation.

	Organ	Substance taken up into the blood	Explanation
(i)		insulin	Insulin is secreted from the organ in response to the change of the blood glucose level.
(ii)		urea	

Ans

2013Q2: C (76%)	2013Q33: A (79%)	2013Q34: C (81%)	2017Q36: A (78%)
2019Q35: A (48%)	2020Q9: D (45%)	2020Q31: D (81%)	2021Q26: B (35%)
2023Q32: B (75%)	2023Q33: C (69%)		

- 2012Q7
- 7 (a) • despite an increase in blood glucose level was detected in Tom’s blood, his insulin level was lower than that of the healthy person (1) (1)
- and the insulin level only showed little change (1) (1)
- this showed that Tom failed to produce the normal amount of insulin (1) (1)
- therefore, Tom suffered from insulin-dependent diabetes (1) (1)
- (b) • with insufficient insulin, his body cells would not take up extra glucose from the blood as efficiently as the healthy person (1) (1)
- as a result, the blood glucose concentration rised to a higher level (1) after glucose consumption (1)
- and remained high for a longer time / decreased slower than the healthy person (1) (1)
- (c) • by injection of insulin / aerosal spray of insulin applied to nasal cavity (1) (1)
- 8 marks

- 2015Q8
- 8 (a) • type I diabetes / insulin dependent diabetes mellitus (1) (1)
- (b) (i) • Lisa’s pancreas failed to secrete enough insulin (1) (1)
- with insufficient stimulation by insulin, there is a reduced uptake of glucose from blood by her body cells for respiration / liver cells for conversion into glycogen (1) (3)
- hence, her blood glucose level dropped very slowly / remained at high level / exceeded the normal range (1) even after 12 hours of fasting
- (b) (ii) • Without the inhibitory effect of insulin, Lisa’s pancreas secretes a large amount of glucagon (1) (3)
- high level of glucagon stimulates liver cells (1)
- to promote the conversion of glucose from glycogen / amino acids (1) maintaining to a high glucose level in her blood
- (c) • having frequent meals but in small portions (1) (1)
- avoid food which elevates blood glucose level in a short time (1) such as sugars and carbohydrates (2)
- 9 marks

2018Q7

- 7 (a) • digestion of foods containing carbohydrates to form glucose (1) (2)
 • absorption of glucose from the small intestine into the blood (1)
- (b) (i) • insulin (1)
- (ii) • it stimulates the body cells and liver cells to take up more glucose from blood (1)
 • increases respiration in body cells to consume glucose (1) (3)
 • it stimulates the conversion of glucose to glycogen by the liver / muscle cells (1)
- (iii) • has initial basal value and drops back to basal value at the end (1), effect lags behind (1) (2)
- (c) • low GI food will lead to small fluctuations in blood glucose level (1)
 • the chance of having too high blood glucose level / glucose appearing in urine is reduced (1) (2)

10 marks

2022Q6b

- 6 (b) • during jogging, muscles continue to take up glucose from blood (1) to release energy via respiration to support muscle contraction
 • the blood glucose level will drop and stimulate the pancreas to secrete more glucagon and less insulin (1)
 • this hormone combination stimulates the liver cells to convert stored glycogen to glucose (1) (4)
 • and release into the blood, thereby increase the blood glucose level back to normal / maintain the normal blood glucose level (1)

2022Q11

11 Concept for mark award:

Managing the blood glucose level in diabetics (max.3):

- as compared with normal diets, the consumption of ketogenic diets will not lead to a sharp rise in blood glucose level / result in a relatively stable blood glucose / (1) because
 - the ketogenic diet is low in carbohydrates, this will end up with a smaller amount of glucose after digestion and absorption (1)
 - after metabolism, only a small amount of glucose will be obtained from the fat of the ketogenic diet (1) max. 3
- diabetics cannot adequately regulate the blood glucose level on their own / cannot produce sufficient insulin to lower the blood glucose level after meal / cannot adequately respond to insulin (1), after consumption of ketogenic diet, the blood glucose level is still manageable to the diabetics / glucose will not appear in the urine of the diabetics (1)

Concept of energy balance for weight management (max. 4):

- to lose weight, the daily energy intake should be lower than the daily energy expenditure / energy deficit (1), when energy intake is lower than energy expenditure, the body will mobilise / use the energy reserve, such as glycogen stored in liver and muscles as energy source (1)
- ketogenic diet is low in carbohydrates, the body is forced to use stored glycogen as energy source (1), if glycogen is insufficient / depleted, the body will use the stored fat instead (1) max. 4
- as the ketogenic diet has a high fat content, the energy intake may be higher than the normal diet (1) because fat stores more energy than the same mass of proteins and carbohydrates (1)

Other health concerns (max. 2): (Any **two**, accept other reasonable answers)

- the high-fat content may result in obesity or blockage of blood vessels due to high fat / triglyceride / cholesterol levels in blood which may result in cardiovascular diseases / stroke (1)
- brain cells use mainly glucose as the energy source, a low carbohydrate intake may result in a low blood glucose supply, leading to dizziness (1) max. 2
- consuming a ketogenic diet will deplete the glycogen store, people may feel tired easily (1)

Effective communication (0-3)

max. 3

12 marks

2024Q4c

	Concept for mark award	Example	Marks
4(c)(i)	correct organ	pancreas	(3)
4(c)(ii)	correct organ	liver	(1)
	explanation in relation to the function of the organ	excess amino acids are deaminated / metabolised to form urea in this organ and transported away	(1)